

Kenechukwu Nwodo

Address: Arlington, VA.

nwodok@vt.edu

Website | LinkedIn | Google Scholar

Summary: Computer Engineering Ph.D. Candidate with research experience in agentic AI for network defense and LLM-driven vulnerability management, with industry experience in mobile security.

EDUCATION

- **Virginia Tech** Alexandria, VA
Ph.D. in Computer Engineering; GPA: 3.80 08/2020 - (Exp.)12/2026
 - **Advisors:** Dr. Angelos Stavrou and Dr. Haining Wang
 - **Dissertation Title:** Securing Software Systems and Autonomous Network Defense
- **Virginia Tech** Arlington, VA
M.S. in Computer Engineering; GPA: 3.78 08/2020 - 05/2023
- **Rensselaer Polytechnic Institute** Troy, NY
B.S. in Electrical Engineering 08/2016 - 05/2020

RESEARCH EXPERIENCE

- **Network and Systems Security Lab, Virginia Tech** Alexandria, VA
Graduate Research Assistant 08/2020 - Present
 - **Autonomous Cyber Agent Training and Evaluation:** Evaluating heuristic and RL-based Red/Blue agents across a hybrid simulation and emulation infrastructure with randomized network topologies, isolating security objective, chokepoint structure, and topology size as independent drivers of agent strategy, to generate actionable insights for remediation and develop stronger defensive agents capable of adapting to realistic environments.
 - **Software Vulnerability Management:** Reconstructing CVE report descriptions with multiple LLM-developed configurations to facilitate easier automation of vulnerability analysis, mapping vulnerabilities from CVEs to MITRE ATT&CK techniques.
 - **Vulnerability Detection:** Automating the identification of vulnerable libraries and packages from Software Bill of Materials in 5G virtual network functions.
 - **SCADA System/IoT Security:** Detecting and mitigating wireless attacks (DoS, Code Injection) on IoT devices and SCADA systems; with the BEMOSS open-source platform, using a shadow replica.
 - **Embedded Systems Security:** Developing a mutual authentication system with STSAFE for FIDO-like applications.
 - **Sensing for Smart Home Monitoring:** Partnering with the School of Construction to create software and hardware for environmental sensor data collection on Raspberry Pis for deployment in Virginia homes.

INDUSTRIAL EXPERIENCE

- **Quokka (Formerly Kryptowire Inc)** McLean, VA
Research Engineer Intern 06/2022 - 12/2022
 - Designed a novel static analysis tool to automatically generate a software bill of materials of libraries (names + versions) that comprise an Android app when obfuscation is applied, and performed a comparative analysis.
- **Boeing** Seattle, WA
Hardware-Electronics Electrical Engineering Intern 05/2019 - 08/2019
 - Created software to give 3D positional system orientation and height measurements using multiple LIDARs for Boeing planes. Also investigated 4G-LTE as a high bit-rate data communication link.
- **SelfArray** Troy, NY
Computer/Electrical Engineering Intern 06/2018 - 08/2018
 - Designed pattern-detection software using Python and OpenCV to test the accuracy and efficiency of the company's patented self-assembly method.

PUBLICATIONS

- **Kenekchukwu Nwodo**, et al. [Title Withheld, Under Blind Review] “Autonomous Cyber Agent Training and Evaluation Framework” *Under Review at a Security Conference, 2026*.
- **Kenekchukwu Nwodo**, et al. [Title Withheld, Under Blind Review] “Isolating the Environmental Factors That Drive Autonomous Cyber Defense Strategy” *Under Review at a Security Conference, 2026*.
- **Kenekchukwu Nwodo**, Angelos Stavrou. “Software Engineering Reliability in the Era of Generic and Agentic AI” *Chapter accepted for publication in Generative and Agentic AI Reliability: Architectures, Challenges, and Trust for Autonomous Systems, Springer Nature, August 5, 2026*. (Link to Springer page)
- **Kenekchukwu Nwodo**, Sudip Maitra, Shihua Sun, Angelos Stavrou, Haining Wang. “Improving Security Vulnerability Descriptions using LLMs” *Chapter accepted for publication in Generative and Agentic AI Reliability: Architectures, Challenges, and Trust for Autonomous Systems, Springer Nature, August 5, 2026*. (Link to Springer page)
- Sudip Maitra, **Kenekchukwu Nwodo**, Tolga Atalay, Angelos Stavrou, Haining Wang. “Towards Securing Access Control in 5G and Beyond with Zero Trust” *Paper accepted for publication in ACM Conference on Data and Application Security and Privacy (CODASPY), 2026*.
- Shihua Sun, Pragya Sharma, **Kenekchukwu Nwodo**, Angelos Stavrou, Haining Wang. “FedMADE: Robust Federated Learning for Intrusion Detection in IoT Networks Using a Dynamic Aggregation Method” *International Conference on Information Security (ISC), 2024*.
- Shihua Sun, **Kenekchukwu Nwodo**, Shridatt Sugrim, Angelos Stavrou, Haining Wang. “ViTGuard: Attention-aware Detection against Adversarial Examples for Vision Transformer” *Annual Computer Security Applications Conference (ACSAC), 2024*.
- David A Tomassi, **Kenekchukwu Nwodo**, Mohamed Elsabagh. “Libra: Library Identification in Obfuscated Android Apps” *International Conference on Information Security (ISC), 2023*.
- **Kenekchukwu Nwodo**, et al. “Towards Accurate and Reliable Industrial Intrusion Detection Systems Using Shadow Replicas” *MS Thesis, 2023*.

TALKS AND PRESENTATIONS

- **TALK: Improving Security Vulnerability Descriptions using LLMs**
To Present at DEF CON 34 - Blacks In Cybersecurity Village at Las Vegas, NV, 08/08/2026
- **POSTER: Bridging Simulation and Emulation for Autonomous Cyber Agent Training and Evaluation**
To Present at the 35th USENIX Security Symposium 2026 at Baltimore, MD, 08/12/2026

PROFESSIONAL SERVICE

- **Reviewer:** IEEE International Conference on Computing and Machine Intelligence, 2024 & 2025
- **Reviewer:** Commonwealth Cyber Initiative’s Central Virginia Node FY27 Proposals for Research Grant & Workforce Development Grant, 2026

TEACHING EXPERIENCE

- **Intro to Computer Networking - ECE 3564, Virginia Tech:** Graduate Teaching Assistant, 01/2022 - 05/2022
Held weekly office hours, proctored exams, supervised two graders, and delivered a lecture on Public Key Cryptography.

COURSE PROJECTS

- **Advanced Machine Learning, Fall 2021:** Implemented NB and SVM supervised machine learning classifiers to perform Intrusion Detection for Industrial Control System networks.
- **Network Security, Spring 2021:** Demonstrated the key reinstallation attack (WPA2 attack) on vulnerable and protected Android phones.
- **Network Architecture and Protocols, Fall 2020:** Performed a comparative analysis on 4G-LTE and 5G, for an islanded microgrid network.
- **CAPSTONE, Fall 2019:** Developed a noise dosimeter with email notifications for a server utilizing Django.

SKILLS & AWARDS

- **Research Skills:** Reinforcement Learning, Simulation, Emulation, MITRE ATT&CK, Prompt Engineering, Vulnerability Detection, Static Analysis, Android Library Identification, Network Hardening, SCADA/IoT Security, Traffic Analysis, ML Applications
- **Relevant Courses:** Network Security, Fundamentals of InfoSec, Network Architecture and Protocols, Advanced Machine Learning, System and Software Security, Optimization Techniques, Object Oriented Programming
- **Programming Languages, Tools, and Operating Systems:** Python, Java, C, Linux (Fedora, Ubuntu, Kali, Raspbian), GPU Servers, vLLM, Ollama, LangChain, Prompt Engineering, PyTorch, TensorFlow, HuggingFace, Git, Docker, Apache Cassandra
- **Affiliations:** IEEE Member, ACM Member, CEISCA (*2024 - Present*), iMentor (CCS *2020*)
- **Grants:** USENIX Security Grants for Black Computer Science Students (*2023*), GREPSEC VI (*2023*)
- **Awards:** Davenport Fellowship by ECE, Virginia Tech (*Spring 2026*)